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AlphaNET/PC® 3.2

Release Notes and Installation Guide

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ALPHANET/PC 3.2 RELEASE NOTES AND INSTALLATION GUIDE

The AlphaNET/PC products allow you to connect an IBM-compatible PC or workstation to an Alpha Micro Ethernet network that uses the AlphaNET communications product. At this time, the major component of AlphaNET/PC is the ETHERNDV PC protocol server. These release notes discuss how to install and use ETHERNDV.

ETHERNDV represents both a step forward and a change in philosophy for Alpha Micro networking products. As opposed to the single-driver/single-Ethernet card world of the WD driver, ETHERNDV moves your network into the modern world of multiple-protocol networks. Because of the significant enhancements in the ETHERNDV product, the WD driver is no longer supported. In fact, the greater demands of the Windows environment *require* you use ETHERNDV instead of the WD driver when using Windows or Windows for Workgroups.

INTRODUCTION TO ETHERNDV

The ETHERNDV program is a PC protocol server. It allows a PC running a network application such as AlphaLAN[1] to connect over an Ethernet cable to an AMOS computer running AlphaNET.

Before the advent of ETHERNDV, PCs and workstations communicating with AMOS computers over Ethernet used the WD driver, which works only for Western Digital WD8003E-chip compatible PC Ethernet cards, limiting the number of Ethernet cards that could be used. Also, the WD driver only supports a single protocol—AlphaNET. To use a non-AlphaNET protocol, you had to reboot your PC without the WD driver.

ETHERNDV allows AlphaNET to run on IBM PC compatible computers configured with one of many different Ethernet adapter cards and allows AlphaNET and other protocols (such as Novell NetWare) to run at the same time on the same Ethernet card.

The way ETHERNDV accomplishes this is that it does not speak directly to the Ethernet card in the PC, but to a “packet driver.” Because ETHERNDV interfaces with a packet driver rather than the Ethernet card itself, AlphaNET can run with every card for which a packet driver has been written without Alpha Micro writing a separate AlphaNET driver

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for each card. Because a large number of packet drivers have been written, this significantly expands the number of Ethernet cards you can use.

ETHERNDV can be removed from memory when not needed to free memory for other applications. Optionally, the packet driver can be removed as well if it is not being used by another protocol.

COMPATIBILITY

AlphaNET/PC 3.2 requires that your PC be running MS-DOS 5.0 or later and AlphaLAN® 4.3 or later, and AlphaNET 2.4 or later on your AMOS computer(s). AlphaNET/PC is compatible with Microsoft Windows 3.1 as well as Windows for Workgroups.

The older WD driver is no longer supported. Any sites using WD.EXE should move to the ETHERNDV protocol server in AlphaNET/PC 3.2.

A copy of AlphaNET/PC is included with each copy of AlphaNET purchased for a single AMOS host computer. The number of Type 2 connections available for PCs and Alpha Micro virtual terminals is controlled by the AlphaNET software license. See the *AlphaNET Release Notes* for details.

NEW FEATURES

The major change in AlphaNET/PC 3.2 is in the user licensing. See the *AlphaNET Release Notes* for details. AlphaNET/PC 3.2 contains the following new utilities that help to maintain the network and troubleshoot problems: NETSPD and NTSTAT.

Verifying Communications: NETSPD

When installing a new network connection, it is sometimes useful to verify that the new node can communication with others. The NETSPD program sends packets to another node and verifies that a response is received. The format of the NETSPD command is:

NETSPD W=*network* G=*group* N=*node* {/*count*}

network is the network number, ***group*** is the group number, and ***node*** is the node number of the node to which you are sending the packet. (These items of information can be found in the PCNET.INI file on the specified node.) ***/count*** is an optional value that tells NETSPD how many packets to send (the default is ten).

Using the /count option gives a more accurate and comprehensive way of measuring your network performance.

If the specified remote computer is not available (not up or not on the network), it will tell you so by immediately returning the message [Remote Not Responding]. It will also show this message if the remote node is running a version of AlphaNET prior to 2.4.

Here is a sample NETSPD display:

NETSPD W=101 G=1 N=1 RETURN

AlphaNET Network Echo, Version X.X(XXX)
Copyright (C) 1994 Alpha Microsystems, Inc.

Check response from Network 101, Group 1, Node 1
Received 10 Responses to 10 Echo Requests

Determining Network Status: NTSTAT

The NTSTAT program provides you with a status report telling you which computers are currently active on the network. To see the status display, enter:

NTSTAT RETURN

A typical status display looks something like this:

```
Systems Available:          net  grp  node
(self)   101,   3,     92   Accounting              PC/AT    DOS Ver 6.20
         101,  21,     60   Sales Department        AMOS/32  2.2C(100)-2
         101,   2      26   Vice President's System PC/AT    DOS Ver 6.20
         101,   3      23   Library/001A-040E       AMOS/32  2.2C(100)-2
         101,   2      12   CEO's System            PC/AT    DOS Ver 6.20
         101,   2,     50   Marketing               AMOS/L   2.2C(100)-2
```

In the display above, your computer is listed first (self). It is on network 101, group 3, and is node 92. **Accounting** is the identifying name assigned to your computer in the SYSNAM.DAT file. The last display on the line tells you what operating system you are running under.

Each computer shown on the NTSTAT display has a physical connection to the network and has been recently active on the network, but is not necessarily active on the network right at this moment.

CONNECTING A PC TO AN ALPHANET NETWORK

You can attach a PC or workstation directly to an AMOS-computer AlphaNET network by simply adding it as another node on the network. This document discusses how to do that.

A more complicated case, of adding an entire non-Ethernet NETBIOS-type PC network to an AlphaNET network through a PC gateway, is covered in the document *Installing a PC Gateway on an AlphaNET Network*, available separately.

Additional Documentation

In addition to the instructions in this document, there are several other documents that can assist you in the configuration and operation of your network:

- *AlphaNET User's Guide*, DSO-00063-00.
- *AlphaNET Installation Guide*, DSO-00064-00.
- *AlphaNET Programmer's Guide*, DSO-00065-00.
- *AM-341 PC Ethernet Controller Installation Instructions*, PDI-00341-00.
- *Installing a PC Gateway on an AlphaNET Network*, DSS-10499-00.
- The owner's manual supplied with your computer.
- The instructions supplied with your terminal emulation/communications package.

Configuring the AMOS Host Computer

Before downloading an AM-920 or PC compatible workstation with AlphaNET/PC, your AMOS host computer or computers must be configured for use with AlphaNET.

The list below contains the basic steps for configuring AlphaNET on AMOS based computers:

1. Install all required network hardware.
2. Download AlphaNET software.
3. Enter the product installation code.
4. Create network initialization file.
5. Create network ERSATZ file.
6. Create SYSNAM.DAT file.
7. Add AlphaNET support statements to the system initialization command file.
8. Create NTSER.JIN file.
9. Create VTSER.JIN file.

See the *AlphaNET Installation Guide*, DSO-00064-00, for further details.

Software Installation for PCs and Workstations

AlphaNET/PC files for AM-920 workstations and PCs must be downloaded onto the PC's hard disk drive. This software is provided on both 3-1/2" and 5-1/4" diskettes. AlphaNET/PC 3.2 includes:

- ETHERNDV.EXE.
- Converter program to allow NDIS drivers to be used with AlphaNET.
- Converter program to allow ODI drivers to be used with AlphaNET.
- IPX protocol for packet drivers.
- A number of packet drivers.

Copy the files from the diskette onto your hard disk. Make sure all the files are located in the same directory as on the diskette.

To run any of the programs downloaded from the DOS diskette, you must be in the directory where the files are located. Or, you can add a PATH statement to your AUTOEXEC.BAT file pointing to the directory containing the DOS support files, allowing you to run the programs from any directory.

New packet drivers are continually becoming available. Some of these are for new Ethernet boards, while some are updates to existing packet drivers.

When installing a new Ethernet board, it is always a good idea to check AMTEC+, the Alpha Micro electronic bulletin board, and commercial bulletin boards (e.g., CompuServe forum, Internet anonymous ftp system, etc.) for any new product driver that may be available.

Hardware Configuration

The following sections contain information on configuring and initializing the hardware necessary for connecting one or more workstations directly into an AlphaNET environment. However, before there can be any interaction between the workstations and the computers running AlphaNET, you must purchase and install a terminal emulation/communications package on each workstation.

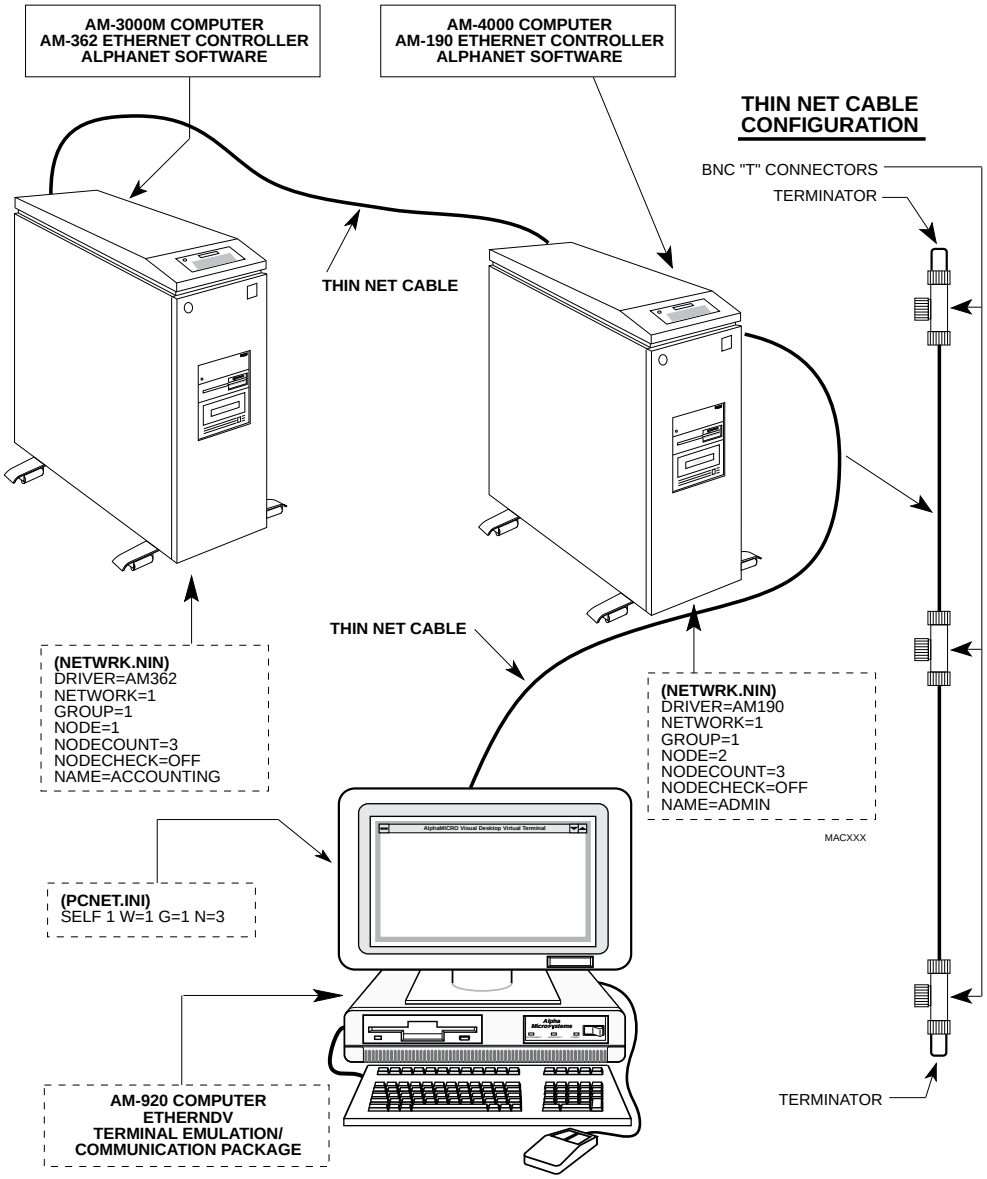


Figure 1. Sample Configuration Using Thin-net Ethernet

Figure 1 shows a basic network with two AMOS-based computers using AlphaNET software. The single AM-920 workstation has an integral Ethernet interface, with rear panel support for both thick and thin Ethernet. The AM-920 also includes AlphaNET/PC ETHERNDV and a terminal emulation/communications package, available separately. The Ethernet controllers in all three computers are linked together using thin net cabling.

Installing ETHERNDV with AlphaNET

This section contains information on installing ETHERNDV for use in the most common situation on Alpha Micro networks: the PC will be using only the AlphaNET protocol. If you want to use the AlphaNET protocol *and* another protocol (such as Novell NetWare), see “Using Multiple Protocols,” later in this document.

In order for a workstation to become active on the AlphaNET network, you must initialize the Ethernet controller by executing a command that loads the appropriate packet driver. The driver used in a direct connection configuration, like the one shown in Figure 1, depends on the Ethernet card being used in the PC. Once the Ethernet controller has been initialized, you can use your terminal emulation/communications program (e.g., AlphaLAN) to control communications between the PC and AMOS computers.

The driver command line takes the form:

DRIVER software-interrupt hardware-interrupt adapter-address

In this example, we will be using the NE1000 driver for a Novell NE-1000 Ethernet adapter at address 0x300 and interrupt 5. (“0x” means hexadecimal.) To interface to the packet driver, we’ll use software interrupt 0x65. The AlphaNET interface will be placed at interrupt 0x62. Additionally, everything in this example will be loaded into low memory—however, most programs (including ETHERNDV.EXE itself) can be loaded into UMB or high memory in order to save space. For information on installing ETHERNDV with protocols other than AlphaNET, see “Using Multiple Protocols,” below.

1. Create the PCNET.INI and SYSNAM.DAT files (discussed below).

2. Copy the NE1000.COM driver from the \DRIVERS directory on the AlphaNET/PC diskette.

3. Execute these commands:

   NE1000 0x65 5 0x300 [RETURN]
   ETHERNDV 0x62 [RETURN]

4. After the DOS prompt reappears, type: **NTSTAT** [RETURN] to ensure you can communicate with other nodes on the network.

5. Load AlphaLAN (i.e., ORBIT.COM) as usual. Be sure to refer to the AlphaLAN documentation for a discussion of the software interrupt vectors used by AlphaLAN to make sure no conflict exists with the interrupts used by ETHERNDV, the packet driver, and the AlphaLAN components. See “Choosing Software Interrupts,” below for more information.

Throughout this document, when we discuss installing ETHERNDV, and then loading the AlphaLAN terminal emulator package, be aware that once AlphaLAN is loaded, the NTSTAT and NETSPD utilities discussed elsewhere in this document cannot be used.

Defining the Network Address: The PCNET.INI File

Each PC on the network has a PCNET.INI file, allocating an AlphaNET address to the PC. When allocating AlphaNET addresses, you must give each computer on the network the same network number (1 through 254), a group number (0 through 254) and a node number (1 through 65530). Each AlphaNET address on the network must be unique. To specify the AlphaNET address, create a PCNET.INI file in the directory where you execute the ETHERNDV command. It must contain SELF 1 W=network-# G=group-# N=node-#. This must be on the first line of the file and must be in upper-case.

For example:

SELF 1 W=101 G=2 N=4

The value following W= is the network number; the value following G= is the group number; and the value following N= is the node number. Simply adjust the values to fit your particular network configuration.

If PCNET.INI isn't found, ETHERNDV allocates its own AlphaNET address. However, since a PCNET.INI file containing a valid network address must exist if you are going to use the node for device redirection (e.g., printing), make sure you have a valid PCNET.INI file.

In the network structure shown in Figure 1, there is a sample PCNET.INI file shown with the AM-920. Note how this file compares to the network initialization files for the AMOS computers. In this simple network configuration, all three computers share the same network and group numbers; only the node numbers are different. This is an important point—computers attached together in a local area network usually share the same network numbers, but each computer must have its own unique group/node number combination. For more information on network initialization files, see the *AlphaNET Installation Guide*.

Defining the Node Name: The SYSNAM.DAT File

The SYSNAM.DAT file contains the name of the computer. This information is displayed when someone uses the network status program, NTSTAT. You can use this file to name each computer in the network so that it may easily be located in case of a problem with the network. If SYSNAM.DAT can't be found when ETHERNDV is executed, ETHERNDV names the node based upon its Ethernet address (for example: AlphaNET node 00:00:B1:00:03:1D).

The SYSNAM.DAT file you create should contain a single line of data limited to a length of 30 characters, and should be located in the same directory you run ETHERNDV from.

Choosing Software Interrupts

ETHERNDV, packet drivers, and network applications (such as AlphaLAN), use PC software interrupts to communicate with each other. These interrupts may be used by other applications, so there is always some potential for conflicts. However, here are some guidelines on values that will work in most cases.

Following are the default and recommended values used by ETHERNDV and various applications. Note that these values may change in the future, so check the current documentation for the application you are using for information on the interrupts used by that package. Values are in hexadecimal.

Module	Default Interrupt	Recommended Interrupt	Range of Interrupts	Function of Module
Packet Drivers	none	65	60-66	Interface to ETHERNDV or other protocol servers.
ETHERNDV	5C	62	5C, 60-66	AlphaNET API interface to applications.
ANDVR	60	60	60-66	*in*Sight/am, AlphaLAN 5.0 for Windows
WINORB	60	60	60-66	AlphaLAN 5.0 for Windows

Using the ETHERNDV Protocol Server

ETHERNDV.EXE is the protocol server for running the AlphaNET protocol on the PC. AlphaNET-aware applications such as AlphaLAN and NTSTAT communicate with the ETHERNDV.EXE program.

To install ETHERNDV.EXE, simply copy it into a directory on the hard disk of the PC (we suggest making an \ALPHANET directory for this purpose).

After you've loaded the packet driver (see the section above for details), you load the AlphaNET protocol by running the ETHERNDV program. The ETHERNDV program then automatically locates the packet driver and starts the AlphaNET protocol. The ETHERNDV command format is:

ETHERNDV {*software-interrupt*}{/*option*}

option may be:

/B	Suppresses node ID broadcasts. (Suppressing ID broadcasts is beneficial in some circumstances—for example, for a network in

	which nodes are connected by modems, and ID broadcasts cause phone-line charges.)
/U	Removes ETHERNDV from memory.
/P	Removes both ETHERNDV and the currently loaded packet driver.

(For more information on the /U and /P options, see “Unloading ETHERNDV and the Packet Driver from Memory,” below.)

The optional ***software-interrupt*** parameter specifies the vector at which to load the AlphaNET protocol. If you omit this parameter, ETHERNDV installs the AlphaNET protocol handler at interrupt 0x5C, the NETBIOS interrupt. However, a number of software packages (especially any version of Microsoft Windows) do not allow the AlphaNET commands to work correctly via interrupt 0x5C because they assume control over the interrupt 0x5C handler. See “Choosing Software Interrupts,” above, for guidelines on choosing the software interrupt.

If you specify a free interrupt from 0x60 to 0x66, ETHERNDV will install itself at that interrupt instead and the AlphaNET interface will work correctly under Windows. AlphaLAN users must be using AlphaLAN version 4.3 or later (ORBIT.COM version 4.3(35) or later and CTRM.COM version 4.3(16) or later).

As an example, if you want to use AlphaNET with Windows using 0x62 as the AlphaNET interrupt, and the packet driver is located at interrupt 0x65, you see something like this displayed:

```
C>ETHERNDV 0x62 [RETURN]
AlphaNET/Packet Driver Interface, Version 2.0(114)
Copyright (C) 1993, Alpha Microsystems Inc.
AlphaNET serial number: 0023-2328
Using packet driver at interrupt 0x65
Waiting for nodelist update...
Driver successfully installed
C>
```

ETHERNDV will run without any additional configuration files; however, ETHERNDV checks for the PCNET.INI and SYSNAM.DAT files as it installs. For information on these files, see above.

If ETHERNDV does not load for any reason, it will set the DOS ERRORLEVEL value to 1 so DOS batch files can display a more obvious error message. See your DOS User’s Guide for details on using ERRORLEVEL in batch files.

Unloading ETHERNDV and the Packet Driver from Memory

ETHERNDV may be unloaded from memory if it is not being used by a program such as AlphaLAN. To do so, use the /U or /P switch. /U removes only ETHERNDV from memory. /P removes both ETHERNDV and the currently loaded packet driver.

You must specify the software interrupt number currently being used by ETHERNDV when removing it unless you used the default, 0x5C. For example:

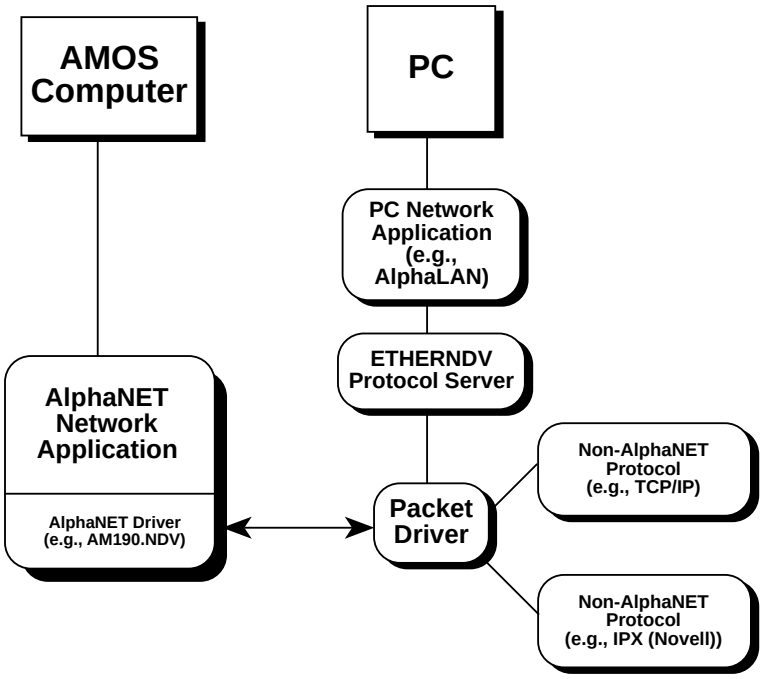
ETHERNDV 0x62/P RETURN

TECHNICAL NOTES ON ETHERNET PACKETS AND DRIVERS

What is a Packet Driver?

A packet driver (originally developed by Clarkson University) is a piece of software that interfaces between Ethernet hardware and various protocol programs. The packet driver specification was written to overcome the limitation of being able to run only one protocol on an Ethernet card at a time (i.e., only AlphaNET or NetWare) and to allow many different Ethernet adapters to be supported. (Before ETHERNDV 3.1, only the Western Digital WD8003E card was supported.) The interface between protocol programs and the packet driver is via a software interrupt. The PC allows 255 different software interrupts, with interrupt numbers 0x60 through 0x66 (0xnn means hex) available for user applications. Almost all packet-driver-aware programs use interrupt 0x65, but if that one is used by other software, you can always specify another interrupt. Remember these numbers are software interrupts and have no relation to hardware interrupts used by various boards in the computer. There are many packet drivers available. For a list and installation instructions read the INSTALL.DOC file in the \DRIVERS directory on the AlphaNET/PC diskette. If your Ethernet board is not listed, check with your Ethernet board vendor to see if they have a packet driver available. For more information on Ethernet standards and packets, see the rest of this section.

Just a word of warning—packet drivers are known by a number of different terms, including “Clarkson drivers,” “Packet drivers,” “FTP Packet drivers,” and “Cynwr Packet Drivers.” All are basically the same thing, but from different sources.



AlphaNET - ETHERNDV Communication

Ethernet Drivers

The following sections give information on various Ethernet drivers, and some guidelines on when to use them.

NDIS Drivers

NDIS drivers perform the same function as packet drivers, but are written to a software interface developed by Microsoft Corporation. These drivers are incompatible with packet drivers at the software level. However, supplied with ALPHANET/PC is a converter program that allows NDIS drivers to be used with AlphaNET. Currently, there are more NDIS drivers available than packet drivers—however, not by many. NDIS drivers occupy more memory than packet drivers, but support different protocol types than packet drivers. Generally speaking, if you can use a packet driver rather than a NDIS driver, use the packet driver as it's faster and uses less memory.

ODI Drivers

ODI drivers are also similar to packet drivers, but conform to a specification written by Novell. ODI drivers are often used for NetWare 386 and by NetWare Lite. Supplied with AlphaNET/PC 3.2 is a converter program, ODIPKT2, that allows ODI drivers to be used with AlphaNET. For an example of how to use ODIPKT2, see "AlphaNET and Novell NetWare (802.3 Packets)," below.

A Note on Ethernet Packets: DIX and 802.3

The physical Ethernet cable supports two types of Ethernet packets. Ethernet II packets are documented in the original Ethernet specification published by DEC, Intel and Xerox (they are also known as DIX packets). The other type, 802.3, conforms to speci-

fication 802.3 published by the IEEE. The difference between the two in simple terms is that DIX packets contain a protocol type at the start of the packet whereas 802.3 packets contain a length field.

Generally speaking, DIX packets are used by multi-user computers (such as UNIX, DEC, and Alpha Micro computers) and 802.3 packets are used for PC-based network operating computers. Because the packet driver specification was written at a university, where mostly multi-user computers (typically UNIX computers running TCP/IP) were in use, the specification only supports DIX packets.

If you only want to use DIX based protocols (AlphaNET and TCP/IP, typically) you won't have a problem. However, if you use Novell NetWare you must either reconfigure your server and all workstations to use DIX packets or use the ODI to packet converter to allow use of both Ethernet packet types.

If your network uses NDIS drivers (i.e., LAN Manager, Windows for Workgroups, 3COM, etc.), you don't have to worry about 802.3 and DIX packets. The NDIS specification is able to handle both types and you would typically use the packet converter only for DIX based protocols such as AlphaNET and TCP/IP.

USING MULTIPLE PROTOCOLS

In the following section, details are given on installing ETHERNDV with a number of different non-AlphaNET protocols that will be used in an AlphaNET network. In these examples, we will be using a Novell NE-1000 Ethernet adapter at address 0x300 and interrupt 5. To interface to the packet driver, we'll use software interrupt 0x65. The AlphaNET interface will be placed at interrupt 0x62. Additionally, everything in these examples will be loaded into low memory; however, most programs (including ETHERNDV.EXE itself) can be loaded into UMB or high memory in order to save space.

AlphaNET and PC/TCP from FTP Software Inc.

Before you start, ensure that you are using the generic version of PC/TCP. This is the version that communicates via packet drivers.

1. Load PC/TCP, following installation instructions supplied with PC/TCP. Ensure that the network connection is working by using the PC/TCP package.

2. Create the PCNET.INI and SYSNAM.DAT files.

3. Execute this command:

   ETHERNDV 0x62 RETURN

4. After the DOS prompt reappears, type: **NTSTAT** RETURN to ensure you can communicate with other nodes on the network.

5. Load AlphaLAN (i.e., ORBIT.COM) as usual. Be sure to refer to the AlphaLAN documentation for a discussion of the software interrupt vectors used by AlphaLAN to make sure no conflict exists with the interrupts used by ETHERNDV, the packet driver, and the AlphaLAN components. See “Choosing Software Interrupts,” above for more information.

AlphaNET and Novell NetWare (DIX Packets)

These instructions assume you are using NetWare version 2.2 or 3.11; for instructions for earlier versions, consult your NetWare administrator.

1. Create the PCNET.INI and SYSNAM.DAT files.

2. Copy the NE1000.COM driver from the \DRIVERS directory on the AlphaNET/PC diskette.

3. Copy the contents of the \NOVELL directory on the AlphaNET/PC diskette to the root directory of the NetWare WSGEN diskette.

4. Log into the drive containing the WSGEN diskette and use WSGEN as usual. Choose the Packet driver - find interrupt option when the list of adapters appears. Follow the normal steps to generate IPX.COM. When done, copy IPX.COM and NETX.COM from the root drive on the WSGEN diskette to your hard drive.

5. Execute these commands on the hard drive:

   NE1000 0x65 5 0x300 [RETURN]
   IPX [RETURN]
   NETX [RETURN]
   ETHERNDV 0x62 [RETURN]

   When you execute the NETX command, you should see the name of the server displayed.

6. After the DOS prompt reappears, type: **NTSTAT** [RETURN] to ensure you can communicate with other nodes on the network.

7. Load AlphaLAN (i.e., ORBIT.COM) as usual. Be sure to refer to the AlphaLAN documentation for a discussion of the software interrupt vectors used by AlphaLAN to make sure no conflict exists with the interrupts used by ETHERNDV, the packet driver, and the AlphaLAN components. See “Choosing Software Interrupts,” above for more information.

AlphaNET and Novell NetWare (802.3 Packets)

These instructions assume you are using NetWare version 2.2 or 3.11; for instructions for earlier versions, consult your NetWare administrator.

1. Create the PCNET.INI and SYSNAM.DAT files.

2. Install NetWare using ODI drivers and NET.CFG. You will use the ODI to packet driver converter, ODIPKT2.

3. Execute these commands on the hard drive:

   LSL [RETURN]
   ODIPKT2 *n* 101 [RETURN]
   IPX [RETURN]
   NETX [RETURN]
   ETHERNDV 0x62 [RETURN]

   where ***n*** is the Ethernet-II protocol entry in NET.CFG.

4. When you execute the NETX command, you should see the name of the server displayed.

5. After the DOS prompt reappears, type: **NTSTAT** [RETURN] to ensure you can communicate with other nodes on the network.

6. Load AlphaLAN (i.e., ORBIT.COM) as usual. Be sure to refer to the AlphaLAN documentation for a discussion of the software interrupt vectors used by AlphaLAN to make sure no conflict exists with the interrupts used by ETHERNDV, the packet driver, and the AlphaLAN components. See “Choosing Software Interrupts,” above for more information.

Diskless Operation of NetWare (Both DIX and 802.3 Packets)

Follow the steps for standard NetWare setup documented above (make sure you use the right packet type, 802.3 or DIX).

After you’ve generated IPX.COM and you’re ready to create the boot diskette, make sure you add the -d switch to the NE1000 command line. This stops the packet driver from initializing the hardware until IPX is ready to run—if you don’t add this switch, your workstation will not boot correctly.

1. Build and test the boot diskette as usual. Do NOT include the ETHERNDV.EXE program on the boot diskette nor in the boot image file (created with the DOSGEN program).

2. Ensure that the ETHERNDV program exists on the server. You can have ETHERNDV executed by either AUTOEXEC.BAT or the NetWare login script.

NDIS-based Networks

This category includes the following products: LAN Manager 2.1, Windows for Workgroups 3.1, MS-DOS Connection for Workgroups, and Chameleon TCP/IP.

1. First, install the NDIS based network per the installation instructions supplied with the network package. Ensure the network is operating correctly before loading ETHERNDV.

2. Find the directory where PROTOCOL.INI is located. To find this directory, look at the device=protman.dos line in your CONFIG.SYS file. The directory specified after the /i: switch is the location of your PROTOCOL.INI file.

3. Using a text editor, look for the line(s) in PROTOCOL.INI that contain the word BINDINGS=. Make a note of what comes after the = sign. This is the logical name of the Ethernet driver (in our case it's MS$NE1000).

4. Add the following lines to the end of PROTOCOL.INI:

   ```
   [PKTDRV]
   bindings = MS$NE1000
   intvec = 0x65
   ```

5. Copy the file DIS_PKT.DOS from the \NDIS directory on the AlphaNET/PC diskette to the directory where PROTOCOL.INI is located.

6. Add the following line to your CONFIG.SYS file:

   device=(path to PROTOCOL.INI)\dis_pkt.dos

7. Reboot the computer to load the packet to NDIS converter.

8. Create the PCNET.INI and SYSNAM.DAT files.

9. Execute this command:

   ETHERNDV 0x62 [RETURN]

10. After the DOS prompt reappears, type: **NTSTAT** [RETURN] to ensure you can communicate with other nodes on the network.

11. Load AlphaLAN (i.e., ORBIT.COM) as usual. Be sure to refer to the AlphaLAN documentation for a discussion of the software interrupt vectors used by AlphaLAN to make sure no conflict exists with the interrupts used by ETHERNDV, the packet driver, and the AlphaLAN components. See "Choosing Software Interrupts," above for more information.

NetWare Lite 1.1 or Later

You must use NetWare Lite Version 1.1 or later with ETHERNDV. If you are using version 1.0, you can obtain update software from many sources, including Alpha Micro's AMTEC+ electronic bulletin board.

1. Install NetWare Lite as usual, and ensure the network is operating correctly before loading ETHERNDV.

2. Go to the directory where NetWare Lite is located (almost always C:\NWLITE) and edit the file STARTNET.BAT using a standard text editor. Replace the first three lines:

   LSL
   NE1000
   IPXODI

   with:

   NE1000 -n 0x65 5 0x300
   IPX

3. Copy the file IPX.COM from the \NOVELL directory on the AlphaNET/PC diskette to the C:\NWLITE directory.

4. Reboot the computer and restart NetWare Lite. After the CLIENT and/or SERVER commands have been executed, you should be able to communicate with other nodes on the network.

5. Create the PCNET.INI and SYSNAM.DAT files.

6. Execute the command:

   ETHERNDV [RETURN]

7. After the DOS prompt reappears, type: **NTSTAT** [RETURN] to ensure you can communicate with the other nodes on the network.

8. Load AlphaLAN (i.e., ORBIT.COM) as usual. Be sure to refer to the AlphaLAN documentation for a discussion of the software interrupt vectors used by AlphaLAN to make sure no conflict exists with the interrupts used by ETHERNDV, the packet driver, and the AlphaLAN components. See “Choosing Software Interrupts,” above for more information.

SPECIAL NOTE FOR WINDOWS FOR WORKGROUP 3.1 USERS

When you install Windows for Workgroups 3.1, if you choose “SMC/Western Digital” as your Ethernet card, you must ensure that you are really using a Western Digital or SMC Ethernet card. The SMCMAC.DOS driver installed by WFW ensures that is running on a true Western Digital or SMC card.

If you are using a Western Digital clone board or MEM MAP mode on an AM-920 workstation, you must use the MACWD.DOS driver found in the \NDIS directory on the

AlphaNET/PC diskette. Follow these steps to convert from SMCMAC.DOS to MACWD.DOS: (However, note that AM-920 users are urged to configure their workstations in I/O MAP mode—see the section below.)

Copy MACWD.DOS from the \NDIS directory on the AlphaNET/PC diskette to the directory where PROTOCOL.INI is located.

Using a text editor, edit the CONFIG.SYS file. Change the line defining the SMCMAC.DOS driver to use MACWD.DOS instead.

Using a text editor, edit the PROTOCOL.INI file. Change any occurrences of SMCMAC to MACWD.

Reboot the PC to use the MACWD driver.

SPECIAL NOTE FOR WINDOWS FOR WORKGROUP 3.11 WITH NETWARE USERS

To run Windows for Workgroups 3.11 when NetWare is already installed, select “IPX Support Driver (Monolithic) with NetBIOS Transport (NDIS 3)” under the Networks selection from the Networks Setup menu. You will need the NetWare Windows files (NETWARE.DRV, VIPX.386, etc.) available from Novell.

If NetWare has already been installed as described previously in this document, no AlphaNET-specific action is required to run with Windows for Workgroups.

SPECIAL NOTE FOR AM-920/386SX AND AM-920/486 USERS

We strongly recommend that AM-920 workstations use the NE1000 NDIS driver supplied with both LAN Manager 2.1 and Windows for Workgroups 3.1 with the workstations configured in I/O MAP mode. However, for this to work correctly, 386SX AM-920s will require a hardware modification. Contact your dealer for information on this modification.